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Model and Data Analysis of Two-Settlement Electricity Market with Virtual Bidding

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Abstract— **Systematic nonzero spreads, defined as the differences between day-ahead and real-time prices, are routinely observed in the wholesale electricity markets. Virtual bidding is a financial mechanism which aims to reduce the magnitude of spreads by allowing market participants to arbitrage on the spread. We follow a data-driven approach to develop a two-settlement market model, and consider a game-theoretic setting with virtual bidders as strategic players. We interpret the spread as a measure of the average forecast accuracy of the market and all the virtual bidders. The main results convey the implication that introducing more qualified virtual bidders into the market help the convergence of the spread.**

I. Introduction

The wholesale electricity markets in the United States are mainly two-settlement markets, each of which consists of a day-ahead (DA) forward market and a real-time (RT) spot market. In the DA market, market participants submit supply and demand bids, and then the independent system operator (ISO) establishes a generation and load schedule with DA locational marginal prices (LMPs) for each of the 24 hours on the next day. In the RT market, market participants buy and sell electricity to balance the differences between day-ahead commitments and the actual load and generation, which establishes the RT LMPs.

In an ideal two-settlement market, the DA LMP equals the expected RT LMP by the no-arbitrage condition. However, this does not hold in the real markets. Systematic nonzero spreads, defined as the differences between DA and RT LMPs, are routinely observed. The statistics of spreads vary by hour and season. When the spread is zero, the market is said to be efficient.

Virtual bidding is a financial mechanism which allows outside financial entities to arbitrage on the spread, without physically consuming or producing power. The purpose of the mechanism is to improve the market efficiency by reducing the magnitude (i.e., absolute value) of the spread. There are two types of virtual bids. A virtual supply bid, called an incremental (INC) bid, is to sell power in the DA market, with the obligation to buy back the same amount of power in the RT market. On the contrary, a virtual demand bid, called a decremental (DEC) bid, is to buy power in the DA market, with the obligation to sell back the same amount of power in the RT market. For example, when a virtual bidder expects that the DA LMP is greater than the RT LMP, it submits an INC bid to make a profit. Such an INC bid increases the supply in the DA market so that the DA LMP decreases. Likewise, the INC bid increases the demand in the RT market so that the RT LMP increases. Consequently, the magnitude of the spread is reduced. For this reason, the mechanism is also called convergence bidding.

Virtual bidding has been implemented in the wholesale electricity markets in the United States. The effectiveness of virtual bidding is acknowledged by the fact that the average spread is closer to zero after its implementation in the two-settlement market. Extensive data analysis has been done in the literature [1]–[6], for different markets and time horizons. The recent literature investigates the market efficiency by testing the existence of profitable trading strategies [7], [8]. A fault with virtual bidding is presented in [9]. They show that virtual bidding cannot handle intra-hour ramping constraints, so that the system cost remains high while the spread converges.

However, there lacks a theoretical model to explain the settlement of DA and RT LMPs (and hence the nonzero spread), and to address the question how and when the virtual bidding mechanism works. Moreover, the two-settlement models in the literature [10]–[12] fail to capture some key features of the real markets. For example, the market operation requires that enough generation capacity be scheduled in the DA market to meet the forecast demand. Consequently, the DA demand almost amounts to the total demand. This is widely different from the financial markets.

The goal of our work is to understand the interactions between DA and RT markets, with the virtual bidding mechanism implemented. Inspired by the results of data analysis on the PJM market, we develop a two-settlement market model, based on which we consider virtual bidders as strategic players, and conduct an equilibrium analysis to illustrate the theory of virtual bidding.

The paper is organized as follows. In Section II, we focus on the PJM market and present the descriptive statistics of the historical data. In Section III, we formulate a two-settlement market model that captures the features specific to the wholesale electricity market. We then study the theory of virtual bidding in Section IV, following a game-theoretic approach. Section V concludes the paper.

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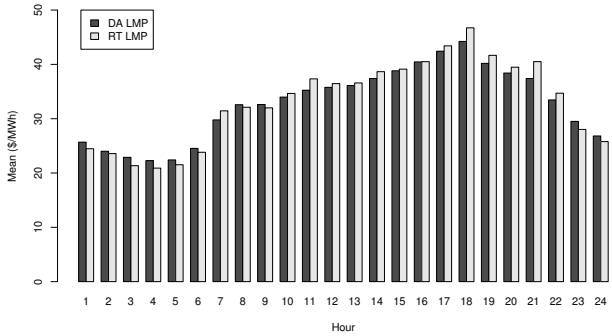

Fig. 1. Mean of DA and RT LMPs by hour, PJM, 2012.

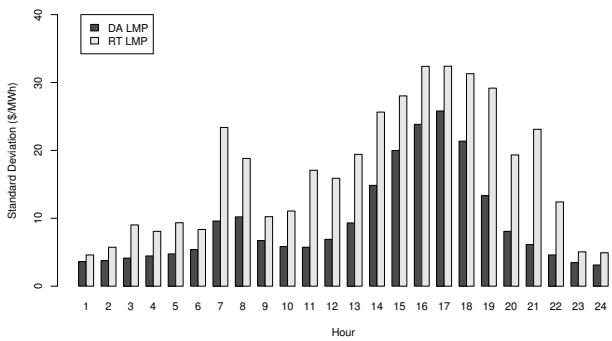

Fig. 2. Standard deviation of DA and RT LMPs by hour, PJM, 2012.

II. Descriptive Statistics of the Historical Data of the PJM Market

In this section, we examine the historical data of the PJM market. The DA and RT LMPs are the weighted average in the PJM region. Fig. 1 presents the mean of DA and RT LMPs by hour in 2012. On average, we have

$$\text{spread} = \text{DA LMP} - \text{RT LMP} \approx 0.$$

The standard deviation of DA and RT LMPs in the same period is shown in Fig. 2. It can be seen that RT LMPs are more volatile than DA LMPs.

Fig. 3 plots the time series of spreads for a particular hour. Spreads are large positive when price spikes occur in the DA market, and large negative when price spikes occur in the RT market. In general, spreads are hard to forecast. The distribution of spreads is heavy-tailed and left-skewed, as shown in Fig. 4. Fig. 5 presents the mean and standard deviation of spreads by hour. The spread is statistically significantly different from zero in 10 out of the 24 hours. Moreover, the mean is negatively correlated with the standard deviation.

Now we study the demand data. Subtracting the wind generation from the load, we obtain the net load, which we also call the total demand:

$$\text{total demand} = \text{net load} = \text{load} - \text{wind generation}.$$

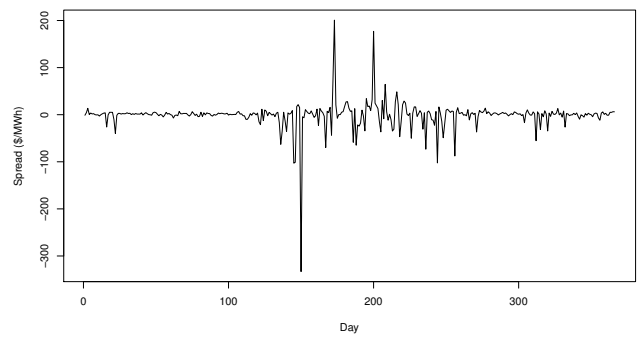

Fig. 3. Time series of spreads, PJM, 2012, Hour 17.

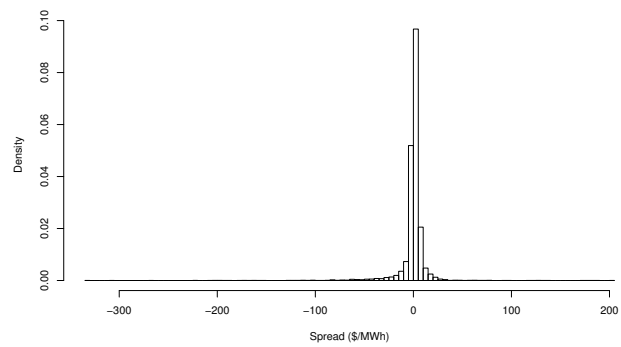

Fig. 4. Histogram of spreads, PJM, 2012.

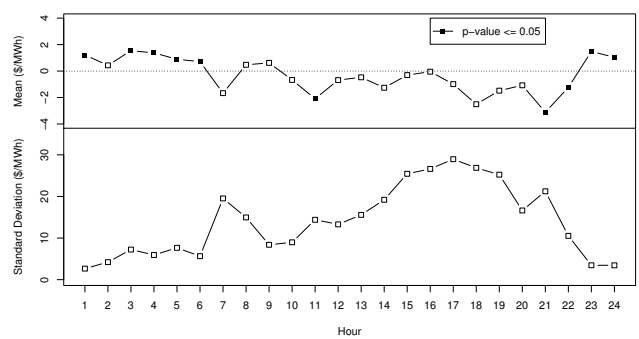

Fig. 5. Mean and standard deviation of spreads by hour, PJM, 2012.

On the other hand, the total demand is the sum of the DA demand and the RT demand:

$$\text{total demand} = \text{DA demand} + \text{RT demand},$$

where the RT demand, which can be both positive and negative, balances the deviation of the actual demand from the DA commitments. Fig. 6 plots the hourly average load, total demand and DA demand. It can be seen that the DA demand almost amounts to the total demand, since the market operation requires that enough generation capacity be scheduled in the DA market to meet the forecast demand.

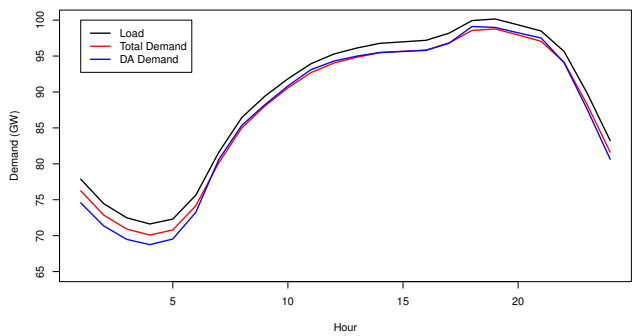

Fig. 6. Mean of load, total demand and DA demand by hour, PJM, 2012.

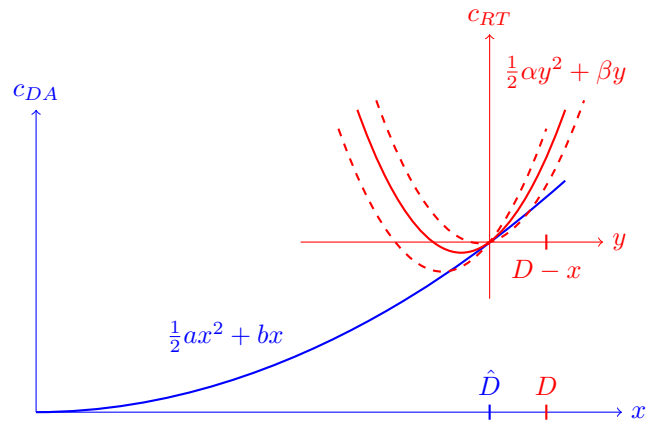

Fig. 7. The two-settlement wholesale electricity market model.

III. Two-Settlement Market Model

The proposed model aims to characterize the relationship between supply functions, demand, prices and forecast errors in the interrelated DA and RT markets. It captures the following features:

1) We assume quadratic cost functions in both the DA and RT markets. Accordingly, the supply functions are affine.
2) The market, as an aggregate of suppliers and demanders, makes forecasts about supply functions and demand, with possible forecast errors. The supply function should be understood in a general sense, which takes into account the effect of transmission congestion, and the contingencies of generation outage, transmission failure, transmission outage, and insufficient ramping capacity.
3) The DA market clears in a deterministic manner given the forecasts. In particular, most of the supply and demand is settled in the DA market, while the RT market functions as a balancing market. This assumption is not unreasonable, and indeed reflects the current practice of the wholesale electricity markets, in contrast to the risk-limiting dispatch paradigm proposed recently [13].
4) While the no-arbitrage condition does not necessarily hold ex post, it holds ex ante from the market's point of view. The no-arbitrage condition couples the DA and RT markets, and should be consistent with the market's forecasts.
5) Since only a small portion of generators (i.e., the fast-start generators) can participate in the RT market, the RT supply function is expected to be steeper than the DA supply function.

A. Model

The two-settlement model is shown in Fig. 7. Let x be the DA demand, and y be the RT demand. Define the cost function of the DA market as

$$c_{DA}(x) = \frac{1}{2}ax^2 + bx,$$

and the cost function of the RT market as

$$c_{RT}(y) = \frac{1}{2}\alpha y^2 + \beta y,$$

where $0 < a < \alpha$. Accordingly, the supply functions are given by

$$\pi_{DA}(x) = ax + b$$

and

$$\pi_{RT}(y) = \alpha y + \beta,$$

where π_{DA} and π_{RT} are the DA and RT LMPs, respectively.

The uncertainties of the system reside on both the demand and supply sides. In particular, we assume that the total demand D and the coefficient β are random variables, whereas a, b and α are constants. The market makes forecasts about D and β, taking the form

$$\hat{D} = D + \epsilon^D$$

and

$$\hat{\beta} = \beta + \epsilon^\beta,$$

where ϵ^D and ϵ^β are the forecast errors.

The market clears in a deterministic manner given the point estimates $\hat{D}$ and $\hat{\beta}$. In the DA market, the DA demand is settled as

$$x = \hat{D},$$

and the RT demand forecast is

$$\hat{y} = 0.$$

The no-arbitrage condition states that the DA LMP equals the forecast RT LMP:

$$\pi_{DA} = \hat{\pi}_{RT} = \alpha\hat{y} + \hat{\beta} = \hat{\beta}.$$

In the RT market, when D and β are realized, the RT demand is settled as

$$y = D - x = D - \hat{D} = -\epsilon_D.$$

The RT LMP is given by

$$\pi_{RT} = \alpha y + \beta = -\alpha\epsilon^D + \hat{\beta} - \epsilon^\beta.$$

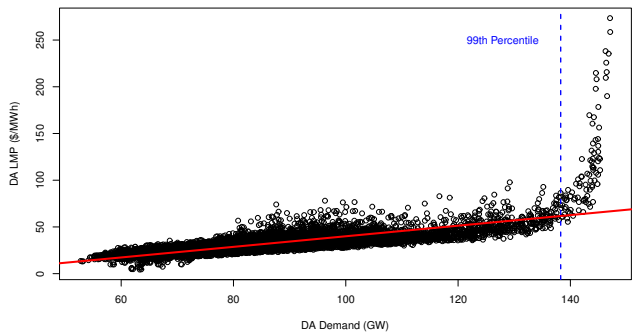

Fig. 8. DA LMP versus DA demand, PJM, 2012.

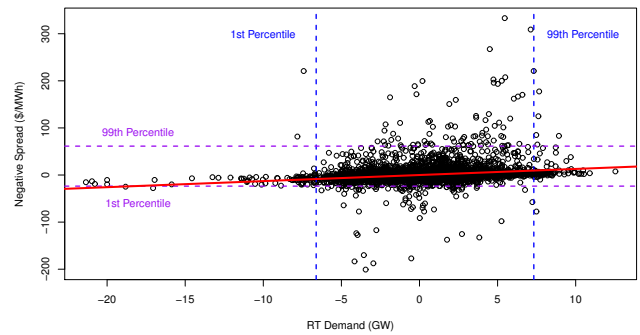

Fig. 9. Negative spread versus RT demand, PJM, 2012.

TABLE I

RESULTS FROM REGRESSION $\pi_{DA} = ax + b + \epsilon$

	Estimate	Standard Error	p-value	R^2
a	0.5667	0.0058	< 0.001	0.5193
b	-16.6607	0.5172	< 0.001	

TABLE II

RESULTS FROM REGRESSION $\pi_{RT} = \alpha y + \gamma \pi_{DA} + h + \epsilon$

	Estimate	Standard Error	p-value	R^2
α	1.2865	0.0626	< 0.001	
γ	0.9915	0.0129	< 0.001	0.4058
h	0.2670	0.4589	0.561	

Therefore, the spread is given by

$$s = \pi_{DA} - \pi_{RT} = \hat{\beta} - (-\alpha\epsilon^D + \hat{\beta} - \epsilon^\beta) = \alpha\epsilon^D + \epsilon^\beta. \quad (1)$$

The interpretation is that the spread measures the forecast accuracy of the market.

B. Empirical Validation

We investigate how the proposed model fits the real market data. Fig. 8 is a scatterplot of DA LMP versus DA demand. As the model suggests, the DA LMP is approximately linear in the DA demand, except when the DA demand is extremely high, possibly due to the transmission congestion. Table I reports the results from the linear regression of the DA LMP on the DA demand. Both the DA demand coefficient and the intercept are statistically significant.

For the RT market, the model suggests

$$\pi_{RT} = \alpha y + \pi_{DA} - \epsilon^\beta.$$

We do a multiple linear regression of the RT LMP on the RT demand and the DA LMP, and report the results in Table II. As expected, the DA LMP coefficient γ is very close to 1, the RT demand coefficient α is greater than the DA demand coefficient a in the previous regression, and the intercept h is not statistically significant. Fig. 9 is a scatterplot of the negative spread ($= \pi_{RT} - \pi_{DA}$) versus the RT demand y. It can be seen that ϵ^β is highly volatile, and the data reveals that it is indeed the dominant term in the spread (1).

IV. THEORY OF VIRTUAL BIDDING

To understand how the virtual bidding mechanism works, we extend the two-settlement model to a game-theoretic setting with virtual bidders as strategic players. Through an equilibrium analysis, we provide a theoretical formula of spread, and discuss the practical implications of the results.

A. Game-Theoretic Formulation

Consider N virtual bidders (VBs), indexed by $i = 1, \ldots, N$. Each VB i has its own forecasts about D and β, taking the form

$$\hat{D}_i = D + \epsilon_i^D$$

and

$$\hat{\beta}_i = \beta + \epsilon_i^\beta,$$

where ϵ_i^D and ϵ_i^β are the forecast errors.

Each VB i places a virtual bid x_i in the DA market. When $x_i > 0$, it is a virtual supply bid (INC): VB i sells x_i amount of power in the DA market, with the obligation to buy the same amount of power in the RT market; when $x_i < 0$, it is a virtual demand bid (DEC): VB i buys $|x_i|$ amount of power in the DA market, with the obligation to sell the same amount of power in the RT market.

Given the virtual bids, the DA demand is settled as

$$x = \hat{D} - \sum_j x_j.$$

VB i's forecast about the RT demand is

$$\hat{y}_i = \hat{D}_i - x + \sum_j x_j = \hat{D}_i - \hat{D} + 2\sum_j x_j.$$

VB i's forecast about the RT LMP is

$$\hat{\pi}_{RT} = \alpha\hat{y}_i + \hat{\beta}_i = \alpha\left(\hat{D}_i - \hat{D} + 2\sum_j x_j\right) + \hat{\beta}_i.$$

VB i's forecast about the spread is then given by

$$\hat{s}_i = \pi_{DA} - \hat{\pi}_{RT} = \alpha\left(\hat{D} - \hat{D}_i - 2\sum_j x_j\right) + \hat{\beta} - \hat{\beta}_i.$$

Note that we have implicitly assumed that the market's forecasts $\hat{D}$ and $\hat{\beta} = \pi_{DA}$ are public information.

The profit or loss per unit of virtual bid is the magnitude of the spread, ignoring transaction costs. When the DA LMP is greater than the RT LMP, an INC bid makes a profit; when the opposite is true, a DEC bid makes a profit. That is, a virtual bid is profitable if and only if it has the same sign as the spread. Formally, VB i's utility function is defined as its forecast profit:

$$\begin{aligned} u_i(x_i, x_{-i}) &= \hat{s}_i x_i \\ &= \left(\alpha \left(\hat{D} - \hat{D}_i - 2 \sum_j x_j \right) + \hat{\beta} - \hat{\beta}_i \right) x_i. \end{aligned}$$

B. Equilibrium Analysis

We are now ready to compute the Nash equilibrium $(x_1, \ldots, x_N)$ in which each x_i maximizes VB i's utility, given the others' bids $x_{-i} = (x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_N)$.

The first-order condition of VB i's utility maximization problem is given by

$$\frac{\partial u_i(x_i, x_{-i})}{\partial x_i} = 0,$$

or

$$x_i = -\frac{1}{2} \sum_{j \neq i} x_j + \frac{\hat{D} - \hat{D}_i}{4} + \frac{\hat{\beta} - \hat{\beta}_i}{4\alpha}. \tag{2}$$

This yields a system of equations. Taking the summation of (2) over i, we obtain

$$\sum_j x_j = -\frac{N-1}{2} \sum_j x_j + \frac{N\hat{D} - \sum_j \hat{D}_j}{4} + \frac{N\hat{\beta} - \sum_j \hat{\beta}_j}{4\alpha},$$

or

$$\sum_j x_j = \frac{N(\alpha\hat{D} + \hat{\beta}) - \sum_j (\alpha\hat{D}_j + \hat{\beta}_j)}{2\alpha(N+1)},$$

which can also be expressed as

$$\sum_j x_j = \frac{N(\alpha\epsilon^D + \epsilon^\beta) - \sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta)}{2\alpha(N+1)}. \tag{3}$$

Substituting (3) into (2) yields an alternative expression of x_i:

$$x_i = \frac{\alpha\epsilon^D + \epsilon^\beta + \sum_{j \neq i} (\alpha\epsilon_j^D + \epsilon_j^\beta) - N(\alpha\epsilon_i^D + \epsilon_i^\beta)}{2\alpha(N+1)}. \tag{4}$$

All the x_i's constitute a Nash equilibrium.

Now we compute the realized spread when D and β are realized in the RT market. The realized RT demand is given by

$$y = D - x + \sum_j x_j = -\epsilon^D + 2 \sum_j x_j.$$

The realized RT LMP is given by

$$\pi_{RT} = \alpha y + \beta = -\alpha\epsilon^D + 2\alpha \sum_j x_j + \hat{\beta} - \epsilon^\beta.$$

Therefore, the spread with virtual bidding is given by

$$\begin{aligned} s^{VB} &= \pi_{DA} - \pi_{RT} \\ &= \hat{\beta} - \left(-\alpha\epsilon^D + 2\alpha \sum_j x_j + \hat{\beta} - \epsilon^\beta \right) \\ &= \frac{\alpha\epsilon^D + \epsilon^\beta + \sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta)}{N+1} \\ &= \frac{s + \sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta)}{N+1}, \end{aligned} \tag{5}$$

where s is the spread without virtual bidding, as in (1).

The interpretation of (5) is that the spread with virtual bidding is a measure of the average forecast accuracy of the market and all the virtual bidders. In the following subsection, we discuss the practical implications of the theoretical results.

C. Implications

First, we provide a sufficient condition under which the magnitude of the spread shrinks with virtual bidding.

Proposition 1: If

$$\frac{1}{N} \left| \sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta) \right| < |\alpha\epsilon^D + \epsilon^\beta|,$$

then

$$|s^{VB}| < |s|.$$

Proof: We have

$$\begin{aligned} |s^{VB}| &= \left| \frac{\alpha\epsilon^D + \epsilon^\beta + \sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta)}{N+1} \right| \\ &\leq \frac{|\alpha\epsilon^D + \epsilon^\beta| + |\sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta)|}{N+1} \\ &< \frac{|\alpha\epsilon^D + \epsilon^\beta| + N|\alpha\epsilon^D + \epsilon^\beta|}{N+1} \\ &= |s|. \end{aligned}$$

■

Therefore, the spread converges when the average forecast of the virtual bidders is better than that of the market. Similarly, it can be shown that a sufficient condition of $|s^{VB}| > |s|$ is the following:

$$\frac{1}{N} \left| \sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta) \right| > 2|\alpha\epsilon^D + \epsilon^\beta|.$$

Next, we consider the case in which the virtual bidders have good forecasts so that the relevant law of large numbers holds. We show that the spread converges to zero as the number of virtual bidders goes to infinity.

Proposition 2 (Cournot Theorem for Virtual Bidding): If

$$\frac{1}{N} \sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta) \to 0 \text{ as } N \to \infty,$$

then

$$s^{VB} \to 0 \text{ as } N \to \infty.$$

Proof: Since

$$\frac{s}{N+1} \to 0$$

and

$$\frac{\sum_j (\alpha\epsilon_j^D + \epsilon_j^\beta)}{N+1} \to 0$$

as $N \to \infty$, we have

$$s^{VB} \to 0 \text{ as } N \to \infty.$$

■

The preceding result is analogous to the Cournot theorem in economic theory, which states that as the number of firms goes to infinity, the price converges to the competitive price (i.e., the marginal cost).

Finally, we examine the relationship between spread convergence and profitability of virtual bidders. We show that a virtual bidder makes a positive profit if and only if its participation drives the spread toward zero.

Proposition 3: Let s_{-i}^{VB} be the spread without the participation of VB i:

$$s_{-i}^{VB} = \frac{\alpha\epsilon^D + \epsilon^\beta + \sum_{j\neq i}(\alpha\epsilon_j^D + \epsilon_j^\beta)}{N}.$$

Then $s^{VB}x_i > 0$ if and only if

$$\begin{cases} 0 < s^{VB} < s_{-i}^{VB}, & s_{-i}^{VB} > 0, \\ s_{-i}^{VB} < s^{VB} < 0, & s_{-i}^{VB} < 0. \end{cases} \tag{6}$$

Proof: We have

$$Ns_{-i}^{VB} = \alpha\epsilon^D + \epsilon^\beta + \sum_{j\neq i}(\alpha\epsilon_j^D + \epsilon_j^\beta)$$

and

$$(N+1)s^{VB} = \alpha\epsilon^D + \epsilon^\beta + \sum_j(\alpha\epsilon_j^D + \epsilon_j^\beta).$$

In view of (4), $s^{VB}x_i > 0$ if and only if

$$s^{VB}(Ns_{-i}^{VB} - N((N+1)s^{VB} - Ns_{-i}^{VB})) > 0,$$

or

$$s^{VB}(s_{-i}^{VB} - s^{VB}) > 0,$$

which holds if and only if

$$\begin{cases} 0 < s^{VB} < s_{-i}^{VB}, & s_{-i}^{VB} > 0, \\ s_{-i}^{VB} < s^{VB} < 0, & s_{-i}^{VB} < 0. \end{cases}$$

■

We emphasize that the sufficient and necessary condition states that the participation should drive the spread toward zero, without changing the sign of the spread. Therefore, it is possible that VB i improves the spread:

$$|s^{VB}| < |s_{-i}^{VB}|, \; s^{VB}s_{-i}^{VB} < 0,$$

but makes a loss:

$$s^{VB}x_i < 0.$$

On the other hand, a virtual bidder with accurate forecasts always makes a profit and hence improves the spread.

Corollary 1: If

$$\alpha\epsilon_i^D + \epsilon_i^\beta = 0,$$

then

$$s^{VB}x_i \geq 0.$$

Proof: We have

$$(N+1)s^{VB} - Ns_{-i}^{VB} = 0.$$

Therefore, (6) holds and the result follows. ■

V. Conclusion

In this work, we follow a data-driven approach to develop a two-settlement model of the wholesale electricity market. Based on the proposed model, we formulate a game-theoretic framework to analyze the virtual bidding mechanism. The equilibrium analysis provides a theoretical interpretation of the spread, which measures the average forecast accuracy of the market and all the virtual bidders.

Our study conveys the following practical implications. First, unqualified virtual bidders with poor forecast accuracy must be screened out. Second, introducing more qualified virtual bidders into the market help the convergence of the spread. Third, this mechanism is self-incentivizing: a virtual bidder can make more profit by improving its forecast accuracy, which is also favorable to the market.

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